

Tommaso Giacomello

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EDUCATION

Bocconi University – Milan, Italy

Sep 2026 - Present

PhD in Computer Science

- Research: Transformer models for understanding gene regulation and synthetic biological data generation.
- Core courses: Computer Science, Modern Machine Learning, Bayesian Statistics, Statistical Physics, Stochastic Processes.
- Recipient of PhD scholarship from Fondo Italiano Scienza (FIS) for DECODE project.
- Supervised by Prof. Andrea Tangherloni.

Bocconi University – Milan, Italy

Sep 2024 - Oct 2026

MSc in Artificial Intelligence

GPA: 28.1/30

- Thesis: “*Gene Regulatory Network Inference from Enformer-based models*”.
- Core courses: Bioinformatics, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Computer Vision and Software Engineering.

Bocconi University – Milan, Italy

Sep 2021 - Jul 2024

BSc in Mathematical and Computing Sciences for Artificial Intelligence

Grade: 106/110

- Thesis: “*Enriched Gene Regulatory Network algorithm applied to cancer cell dataset*”.
- Core courses: Algebra, Analysis, Physics, Statistics, Machine Learning, Optimization, Algorithms, Neuroscience.

RESEARCH EXPERIENCE

Oxford University – Oxford, UK

Feb 2026 – May 2026

Visiting Researcher – Weatherall Institute of Molecular Medicine

- Applied pre-trained transformer-based models (Enformer, AlphaGenome) to infer cell-type specific Gene Regulatory Networks from multi-omics epigenomic data (RNA-seq, ATAC-seq, TF ChIP-seq).
- Worked with large-scale datasets and HPC infrastructure (SLURM, GPU-accelerated training).
- Supervised by Prof. Jim Hughes and Simone Riva.

Bocconi University – Milan, Italy

Sep 2024 – Present

Assistant Researcher – Department of Computing Sciences

- Developed machine learning methods for Gene Regulatory Network inference using scRNA-seq and scATAC-seq data.
- Extended the CellOracle framework with enrichment-based methods to improve biological interpretability.
- Awarded three research contracts under the ERC-funded MicroC project.
- Supervised by Prof. Francesca Buffa.

PUBLICATIONS

Giacomello et al. (2026): “*pysigscore: gene signatures scoring across bulk and single-cell transcriptomics*”, (Under Review)

- Developed Python package (pysigscore) for computing signature scores in bulk and scRNA-seq data.
- Extended functionality of the original R library (alebarberis/sigscores) and deployed as a PyPI package.

Giacomello et al. (2025): “*Enrichment Algorithms for Gene Regulatory Network Reconstruction*”, CIBB 2025

PROJECTS

Investigating Public Concerns About AI

NLP Group Project

- Analysed public concerns about AI by building a 20k+ tweet and 500+ article dataset.
- Fine-tuned Twitter-RoBERTa and DistilBERT (~70% F1-score on negative sentiment classification).
- Applied topic modelling (NMF, LDA) to extract structured themes in media framing.

TP53 mutation prediction

AI Lab Group Project

- Built a pipeline for preprocessing and batch correction of CCLE (Cancer Cell Line Encyclopedia) datasets.
- Predicted TP53 mutation status using Random Forest and XGBoost (~81% F1-score), optimized using Optuna library.
- Project work later used by Prof. Salwa Lin in a research grant proposal.

Deep Learning for Sign Language Prediction

Deep Learning Project

- Processed World Level American Sign Language dataset (WLASL), 21k+ videos and 2000+ signs.
- Achieved 60% F1-score on 20 sign classification using 2D-CNN on hand landmarks.

SKILLS

Languages:	Italian (Native), English (C1 Fluent, IELTS 7.5), Spanish (B1 Business, University Exam 27/30).
Programming:	Python, C, R.
AI Libraries:	NumPy, Pandas, PyTorch, XGBoost, LangChain, Scikit-Learn, Optuna, Keras, Captum, Hugging Face.
Bioinformatics:	Scanpy, CellOracle, deepTools.
Tools:	Linux shell, Git, HPC (SLURM), GPU computing.

PRIVACY

I authorize the use of my personal data according to GDPR 679/2016.